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3.④ Central Limit Theorem.

~~1. What is a Sampling Distribution?~~

What it means:

↳ If you take many small samples from any population (even weirdly shaped data).
↳ Calculate the average of each sample
↳ Plot all those averages → they'll form a perfect bell curve (normal distribution).

→ This is Empirical rule Explanation

Why it's cool:

1. Works even if original data isn't normal (e.g. incomes, website visits).
2. Kicks in at about $n = 30$ (sample size of 30+ gives nice bell curves).
3. Explains why polls work (averages opinions become predictable).

Data Science Reality Check:

↳ CLT is less crucial now because we often:
  ↳ Have real data (no need to sample).
  ↳ Use bootstrap methods instead.
  ↳ Work with huge datasets where CLT always applies.

Standard Error – The "Margin of Error" Metric

What it measures:

↳ How much your sample statistic (like an average) bounces around.

Key formula:

Standard Error = (Standard Deviation) / √(sample size)

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practical implications

1. Bigger samples → smaller error (more precise estimates).
2. To cut error in half, quadruple your sample size.
3. The "±" numbers you see in polls (e.g. "60% ± 3%").

Python code

```
import numpy as np
import matplotlib.pyplot as plt

#create a weird non-normal data (exponential + outliers).
data = np.concatenate([
    np.random.exponential(scale=1, size=500),
    np.random.normal(loc=10, scale=1, size=50)])
#Simulate CLT with different sample sizes.
plt.figure(figsize=(12,8))

for i, n in enumerate([1, 5, 30, 100]):
    sample_means = [np.random.choice(data, n).mean()
                    for _ in range(1000)]

    plt.subplot(2, 2, i+1)
    plt.hist(sample_means, bins=30, alpha=0.7)
    plt.title(f"sample size = {n}|nstd Error
              = {np.std(sample_means):.2f}")

    if n==1: plt.xlim(0,12)
plt.tight_layout()
plt.show()
```

→ style confusion

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1) You're simulating

→ 500 values from an exponential distribution (right-skewed, like wait times or survival times).

→ 50 outliers from a normal distribution centered at 10 to mess things up even more.

→ Combined using np.concatenate() → a heavily skewed, non-normal dataset.

2) Setting up a big figure to show 4 subplots

→ This loop is doing sampling simulation to demonstrate the Central Limit Theorem (CLT) visually.

→ looping through different sample sizes.

- n = 1 → tiny sample
- n = 5 → small sample
- n = 30 → decent sample
- n = 100 → big sample.

→ enumerate(...)

↳ It gives you both:

↳ The index i (like 0, 1, 2, 3) → helpful for subplot positions

↳ The actual sample size n (here: 1, 5, 30, 100).

→ sample_means = [... for _ in ...].

↳ ① np.random.choice(data, n)

b) Randomly selects [n] values from the array data (with replacement).

↳ Example: if n = 5, it picks 5 random numbers from data.

② .mean()

→ calculates the average (mean) of these [n] numbers.

③ The result is one sample mean, and we do this 1000 times.

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→ How can we generate 1000 sample from a dataset of just 550 values?

→ The functions np.random.choice(data, n) by default uses replacement, unless you tell it not to.
That means:
1. Each time it selects [n] data points (like 5 or 30 or 100)
2. It can reuse values from the data array.
3. So you're not running out of data - values can repeat in each sample.

(5) plt.subplot(2, 2, i+1) - what it means?
General format:
plt.subplot(n_rows, n_cols, plot_number)
So in this case:
- [2, 2] → You're dividing the entire plot area into a 2×2 grid (4 plots total).
- [i+1] → This is the position of the current subplot (ranging from 1 to 4).

We use [i+1] because enumerate([1, 5, 30, 100]) gives i = 0, 1, 2, 3, but subplots starts counting from 1, not 0.

③ for i, n in enumerate([1, 5, 30, 100])
sample_means = [np.random.choice(data, n).mean() for _ in range(1000)]

① Who maintains fixed sample size and random keeps?

☑ np.random.choice(data, n)
→ Maintains fixed sample size (n)
→ Generate a new random sample each time (with replacement by default).

☑ for _ in range(1000):
→ Repeat the sampling process 1000 times (calls np.random.choice(...) 1000 times)

② Does for _ in range(1000) refill and resample?
→ Yes — each iterations of the loop calls np.random.choice(...) again, so it np.random.choice(...) refills and resample independently each time.

→ Sample size stays fixed (550), and randomness is re introduced every loop.

③ What is happening :

1. Loop over different same sizes : n = 1, 5, 30, 100
2. For each [n], repeat 1000 times.
   - Randomly sample [n] data points (with replacement).
   - Compute their mean.
3) Collect 1000 means → [sample_means].
4) Plot histogram of [sample_means]

④ Result

→ when [n=1] : sample means look like original data.
→ As [n] increases : sample means form a normal-shaped distribution.
→ This is CLT in actions.

Pure CLT math

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed (i.i.d) random variables from a population with:

- Mean μ (known Population Mean)
- Variance σ^2 (known population variance)
- sample size n

Then the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

As $n \to \infty$, the distribution of $\bar{X}_n$ approaches a normal distribution:

$$\bar{X}_n \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right).$$

Example

Let's say the original population is a discrete uniform distribution:

$$x \in \{1, 2, 3, 4, 5\}$$

So, the population has:

- Mean $\mu = \frac{1+2+3+4+5}{5} = 3$
- Variance $\sigma^2 = \frac{1}{5} \sum_{i=1}^{5} (i-3)^2 = 2$

Suppose we take samples of size $n = 4$

Let's take 4 independent values from X:

$$X_1, X_2, X_3, X_4 \sim \text{uniform}(\{1, 2, 3, 4, 5\}).$$

Let $\bar{X}_4 = \frac{X_1 + X_2 + X_3 + X_4}{4}$

Then by CLT: Since the sample size is small but still more than 1, CLT approximately holds.

So, distributions of $\bar{X}_4$ is approximately

$$\bar{X}_4 \sim N(\mu = 3, \sigma^2/n = 2/4 = 0.5)$$

or: $\bar{X}_4 \sim N(3, 0.5)$

~~What~~ How Mathematical CLT and Empirical rule ($n \geq 30$) connected and share relationship?

Core Connections

Mathematically, CLT says.

"As sample size $n \to \infty$, the sampling distribution of the mean becomes normal

Empirically, we say

"Around $n = 30$, their distribution is already close enough to normal for most purpose.

Why $n = 30$?

Not from theory, it's from
- → simulations
- → Experience
- → Observations.

↳ It's a rule of thumb, not a theorem.

The Bridge

The empirical rule is just an approximation of when the mathematical behavior kicks in visibly.